

Q6 - continued

$$r_{\text{primal}} = \epsilon_{\text{abs}} \sqrt{p} + \epsilon_{\text{rel}} \max(\|x\|_2^2, \|z\|_2^2)$$

$$r_{\text{dual}} = \epsilon_{\text{abs}} \sqrt{n} + \epsilon_{\text{rel}} \|-py\|_2^2$$

$$\epsilon_{\text{primal}} = \|x^k - z^k\|_2^2$$

$$\epsilon_{\text{dual}} = \|-p(z^k - z^{k-1})\|_2^2$$

Stop if

$$r_{\text{primal}} \leq \epsilon_{\text{primal}} \cap r_{\text{dual}} \leq \epsilon_{\text{dual}} == \text{True}$$

Else proceed iterations.

Q6 - continued

$$z_i = 0, \quad [-\lambda_1 - y_i - \rho x_i, \lambda_1 - y_i - \rho x_i] \in 0$$

$$\Rightarrow z_i = 0 \quad \text{if} \quad y_i + \rho x_i \in [-\lambda_1, \lambda_1]$$

$$z_i < 0, \quad -\lambda_1 + \lambda_2 z_i - y_i - \rho x_i + \rho z_i = 0$$

$$\Rightarrow z_i^* = \frac{\lambda_1 + y_i + \rho x_i}{\lambda_2 + \rho}, \quad y_i + \rho x_i < -\lambda_1$$

$$z_i > 0, \quad \lambda_1 + \lambda_2 z_i - y_i + \rho x_i - \rho z_i = 0$$

$$\Rightarrow z_i^* = \frac{-\lambda_1 + y_i + \rho x_i}{\lambda_2 + \rho}, \quad y_i + \rho x_i > \lambda_1$$

$$z_i^* = \left(\frac{-y_i + \rho x_i - \lambda_1}{\lambda_2 + \rho} \right)_+ - \left(\frac{-y_i - \rho x_i - \lambda_1}{\lambda_2 + \rho} \right)_+$$

$$\text{where } h(a)_+ = \max(0, a)$$

$$x^{k+1} = (A^T A + \rho I)^{-1} (A^T c + \rho z - y)$$

$$x^{k+1} = (A^T A + \rho I)^{-1} (A^T c + \rho(z^k - u^k))$$

$$z_i^{k+1} = \left(\frac{\rho(x_i^{k+1} + u_i^k) - \lambda_1}{\lambda_2 + \rho} \right)_+ + \left(-\frac{(\rho x_i^{k+1} + u_i^k) - \lambda_1}{\lambda_2 + \rho} \right)_-$$

$$u^{k+1} = u^k + (x^k - z^k)$$

$$\text{Stopping criteria,} \quad A = -B = I \quad C = 0$$

Define a stopping criteria based on relative residual $\epsilon_{\text{relative}}$ and constant error $\epsilon_{\text{absolute}}$

HW1-Q6, Lasso, ADMM $P(\beta) = \alpha \|\beta\|_1 + (1-\alpha) \|\beta\|_2^2$

$$\lambda_1 = \alpha \lambda, \quad \lambda_2 = \lambda(1-\alpha)$$

$$\arg\min_x L_p(x, y, z) = \arg\min_x \frac{1}{2} \|Ax - C\|_2^2 + \lambda_1 \|z\|_1$$

$$= \arg\min_x \frac{1}{2} \|Ax - C\|_2^2 + \lambda_1 \|z\|_1 + \frac{\lambda_2}{2} \|z\|_2^2 + y^T(x - z) + \frac{\rho}{2} \|x - z\|_2^2$$

$$= \arg\min_x \frac{1}{2} (x^T A^T A x - 2x^T A^T C + C^T C) + \frac{\lambda_2}{2} z^T z + y^T x - y^T z + \frac{\rho}{2} (x^T x - 2x^T z + z^T z)$$

$$= \arg\min_x \frac{1}{2} (x^T A^T A x - 2x^T A^T C) + y^T x + \frac{\rho}{2} (x^T x - 2x^T z + z^T z)$$

$$\Rightarrow A^T A x - A^T C + y + \rho(x - z) = 0$$

$$x = (A^T A + \rho I)^{-1} (A^T C + \rho z - y)$$

$$\arg\min_z L_p(x, y, z) =$$

$$= \arg\min_z \frac{1}{2} \|Ax - C\|_2^2 + \lambda_1 \|z\|_1 + \frac{\lambda_2}{2} \|z\|_2^2 + y^T(x - z) + \frac{\rho}{2} \|x - z\|_2^2$$

$$= \arg\min_z \lambda_1 \|z\|_1 + \frac{\lambda_2}{2} z^T z - y^T z + \frac{\rho}{2} (z^T z - 2z^T x)$$

$$\partial \lambda \|z\|_1 = \partial \lambda \sum_{i=1}^p |z_i| = \lambda \sum_{i=1}^p \partial |z_i|$$

$$\frac{\partial L_p(x, y, z)}{\partial z} = \frac{\partial \lambda \|z\|_1}{\partial z} + \nabla_z \left(\frac{\lambda_2}{2} z^T z - y^T z + \frac{\rho}{2} (z^T z - 2z^T x) \right)$$

$$= \frac{\partial \lambda}{\partial z} \|z\|_1 + \lambda_2 z - y + \rho(z - x)$$

$$\frac{\partial L_p(x, y, z)}{\partial z_i} = \begin{cases} -\lambda_1 + \lambda_2 z_i - y_i - \rho x_i + \rho z_i & \text{if } z_i < 0 \\ [-\lambda - (y_i + \rho x_i) \lambda_1 - (y_i + \rho x_i)] & \text{if } z_i = 0 \\ \lambda_1 + \lambda_2 z_i - y_i - \rho x_i + \rho z_i & \text{if } z_i > 0 \end{cases}$$

HW1 Q5

Compute $\lambda_{\max}$ for Lasso estimator

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} L(\beta, \lambda) = \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

$$\partial_{\beta} L(\beta, \lambda) = -\frac{1}{n} X^T (y - X\beta) + \lambda \underbrace{S(\beta)}_{\text{subdifferential}}$$

$$S(\beta_i) = \begin{cases} 1 & \beta_i > 0 \\ -1 & \beta_i < 0 \\ [-1, 1] & \beta_i = 0 \end{cases}$$

$$\partial_{\beta_i} L(\beta, \lambda) = \left[-\frac{1}{n} X^T (y - X\beta) \right]_i + \lambda S(\beta_i)$$

$$\text{KKT}, \hat{\beta}_{\text{lasso}} = 0 \quad 0 \in \partial_{\beta_i} L(0, \lambda) = -\frac{(X^T y)_i}{n} + \lambda S(\beta_i)$$

$$0 = -\frac{(X^T y)_i}{n} + \lambda S(\beta_i)$$

$$|S(\beta_i)| \leq 1$$

$$\left| \frac{(X^T y)_i}{n} \right| = |\lambda S(\beta_i)| \leq \lambda \quad \forall i = 1, 2, \dots, p$$

$$\lambda \geq \max_i \left| \frac{(X^T y)_i}{n} \right| = \frac{1}{n} \|X^T y\|_{\infty}$$

$$\lambda = \frac{1}{n} \|X^T y\|_{\infty}$$

HW1 - Q(4)

Derive expression for ridge estimator & prediction $\hat{y}$, using SVD of X . ~ highly correlated features

$$\beta^R = (X^T X + \lambda I)^{-1} X^T y \quad \hat{y}^R = X(X^T X + \lambda I)^{-1} X^T y$$

Ridge SVD of $X \Rightarrow X = U \Lambda V^T$

$$\begin{aligned} \hat{\beta}^R &= ((U \Lambda V^T)^T U \Lambda V^T + \lambda I)^{-1} (U \Lambda V^T)^T y \\ &= (V \Lambda^2 V^T + \lambda I)^{-1} V \Lambda U^T y \\ &= V (\underbrace{\Lambda^2 + \lambda I})^{-1} \Lambda U^T y \end{aligned}$$

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for conditions $\sigma \approx 0$, $\lambda \neq 0$;

with Ridge regression, due to $\lambda \neq 0$, the error will not have inflated values, that will be close to zero.

$$\text{vs } \hat{\beta}^{LS} = (X^T X)^{-1} X^T y = V(\Lambda^2)^{-1} \Lambda U^T y \quad (\lambda = 0)$$

$$\left[\begin{array}{cccc} \frac{1}{\sigma_1} & & & \\ & \frac{1}{\sigma_2} & & \\ & & \ddots & \\ & & & \frac{1}{\sigma_p} \end{array} \right] \quad \sigma \rightarrow 0$$

If $\sigma \rightarrow 0$, $\frac{1}{\sigma} \uparrow \uparrow$; this cause large deviation

Ridge regression takes care of this due to $\lambda \neq 0$;

$$R \rightarrow \frac{\sigma^2}{\sigma^2 + \lambda}$$

HW1 - Q(3)

Ridge Regression, suppose $y \sim N(X\beta, \sigma^2 I)$
 $\beta \sim N(0, \tau^2 I)$

Calculate the posterior mean and mode (MAP estimator) and show how these relate to ridge regression.

$$\beta^R = \underset{\beta}{\operatorname{argmin}} \|X\beta - y\|_2^2 + \lambda \|\beta\|_2^2$$

$$\begin{aligned} \nabla_{\beta} &= 0 \Rightarrow X^T(X\beta^R - y) + \lambda \beta^R = 0 \\ (X^T X + \lambda I) \beta^R &= X^T y \quad \lambda > 0 \\ \beta^R &= (X^T X + \lambda I)^{-1} X^T y \end{aligned}$$

MAP, G: $\beta^C = \underset{\beta}{\operatorname{argmin}} \max_{\beta} L(y; \beta) \pi(\beta) \quad L(y; \beta) = p(y|\beta)$

(take log) $= \underset{\beta}{\operatorname{argmax}} \log L(\beta) + \log \pi(\beta) \quad \text{taking log}$

$$= \underset{\beta}{\operatorname{argmax}} -\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) - \frac{1}{2\tau^2} \beta^T \beta$$

$$\stackrel{\sigma^2}{\sim} = \underset{\beta}{\operatorname{argmin}} \frac{1}{2\sigma^2} (\beta^T X^T X \beta - 2\beta^T X^T y) + \frac{1}{2\tau^2} \beta^T \beta$$

$$= \underset{\beta}{\operatorname{argmin}} \beta^T \left(X^T X + \frac{\sigma^2}{\tau^2} I \right) \beta - 2\beta^T X^T y$$

(differentiate)

$$\nabla_{\beta} = 0 \quad X^T y = \left(X^T X + \frac{\sigma^2}{\tau^2} I \right) \beta^C$$

$$(X^T X + \lambda I) \beta = X^T y$$

1. continued

$$\nabla_{\lambda} \mathcal{L} = \sum_{i=1}^P \frac{2d^2}{(\lambda+d)^3} + \frac{2V^2\beta^2 d^2 \lambda}{\lambda^2 (\lambda+d)^3}$$

$$= \sum_{i=1}^P \frac{-\lambda^2 2d^2 + 2V^2\beta^2 d^2 \lambda}{\lambda^2 (\lambda+d)^3} = 0$$

$$2V^2\beta^2 d^2 \lambda = \lambda^2 2d^2$$

$$\lambda_{\min} = \frac{\lambda^2}{V^2\beta^2}$$

$$\frac{1}{P} \frac{d^2}{\left(\frac{\lambda^2}{V^2\beta^2} + d^2\right)} + \frac{\lambda^2}{P \lambda^2} \frac{1}{V^2\beta^2} \frac{(V\beta)^2 d^2}{\left(d + \frac{\lambda^2}{V^2\beta^2}\right)^2}$$

$$\frac{1}{P} \frac{d^2 V^2 \beta^2}{\lambda^2 + d^2 V^2 \beta^2} + \frac{\lambda^2}{P \lambda^2 V^2 \beta^2} \frac{(V\beta^2)^2 d^2}{(V^2 \beta^2 d + \lambda^2)} (V^2 \beta^2)^2$$

$$\frac{1}{P} \frac{1}{(V^2 \beta^2 d^2 + \lambda^2)} \left(d^2 V^2 \beta^2 + \frac{\lambda^2}{\lambda^2} (V\beta^2)^2 d^2 (V^2 \beta^2) \right)$$

1. continued

$$\text{MSE}(X\hat{\beta}^R) = \sigma^2 \text{trace} \left(X(X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1} X^T \right) \\ + \lambda^2 \beta^T (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1} \beta$$

$$= \sigma^2 \text{trace} \left(X(X^T X + \lambda I)^{-1} X^T \right)^2 \\ + \lambda^2 \beta^T (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1} \beta$$

$$\left\{ X^T X = V \Lambda V^{-1} \quad V V^{-1} = I \right\}$$

$$= \sigma^2 \text{trace} \left(\left(X V (\Lambda + \lambda I)^{-1} V^{-1} X^T \right)^2 \right)$$

$$+ \lambda^2 \beta^T \left(V (\Lambda + \lambda I)^{-1} V^T V \Lambda V^T V (\Lambda + \lambda I)^{-1} V^T \right) \beta$$

$$= \sigma^2 \text{trace} \left((X V (\Lambda + \lambda I)^{-1} V^{-1} X^T)^2 \right)$$

$$+ \lambda^2 \beta^T V (\Lambda + \lambda I)^{-1} \Lambda (\Lambda + \lambda I)^{-1} V^T \beta$$

$$\text{MSE}(X\hat{\beta}^R) < \text{MSE}(X\hat{\beta}^{LS})$$

$$\frac{1}{\sigma^2 p} \left(\sigma^2 \text{trace} \left[(X V (\Lambda + \lambda I)^{-1} V^{-1} X^T)^2 \right] + \lambda^2 \beta^T V (\Lambda + \lambda I)^{-1} \Lambda (\Lambda + \lambda I)^{-1} V^T \beta \right) < \frac{\sigma^2 p}{\sigma^2 p}$$

$$L = \frac{1}{p} \text{trace} \left[(X V (\Lambda + \lambda I)^{-1} V^{-1} X^T)^2 \right] + \frac{\lambda^2 \beta^T V (\Lambda + \lambda I)^{-1} \Lambda (\Lambda + \lambda I)^{-1} V^T \beta}{\sigma^2 p} < 1$$

$$L < 1$$

$$L = \frac{1}{p} \sum_{i=1}^p \frac{d_i^2}{(\lambda + d_i)^2} + \frac{\lambda^2}{p \sigma^2} \sum_{i=1}^p \frac{(V \beta)_i^2 d_i}{(d_i + \lambda)^2}$$

Take the derivative of L , $\nabla_{\lambda} L = 0$, find $\lambda_{\min}$

HW1 - Theory & Methods

(1) Theory: Ridge Regression. Prove the MSE Existence theorem.

There exists a value of λ for which

$$\text{MSE}(\hat{y} | \hat{\beta}^{\text{Ridge}(\lambda)}) < \text{MSE}(\hat{y} | \hat{\beta}^{\text{LS}}), \quad \hat{y}, X\hat{\beta}$$

$$\text{MSE}(\hat{y}^{\text{LS}}) = \text{MSE}(X\hat{\beta}^{\text{LS}}) = \|\text{Bias}(X\hat{\beta}^{\text{LS}})\|_2^2 + \text{trace}(\text{Var}(X\hat{\beta}^{\text{LS}}))$$

LS unbiased $\Rightarrow$

$$\text{Bias}(X\hat{\beta}^{\text{LS}}) = E[X\hat{\beta}^{\text{LS}} - X\beta] = X[E[\hat{\beta}^{\text{LS}} - \beta]] = X0 = 0$$

$$\begin{aligned}\text{Var}(X\hat{\beta}^{\text{LS}}) &= X \text{Var}(\hat{\beta}^{\text{LS}}) X^T = X \text{Var}((X^T X)^{-1} X^T y) X^T \\ &= X (X^T X)^{-1} X^T \text{Var}(y) X^T = \sigma^2 X (X^T X)^{-1} X^T\end{aligned}$$

$$\begin{aligned}\text{MSE}(\hat{y}^{\text{LS}}) &= \|0\|_2^2 + \text{trace}(\sigma^2 X (X^T X)^{-1} X^T) \\ &= \sigma^2 p\end{aligned}$$

$$\text{MSE}(\hat{y}^{\text{R}})$$

$$\begin{aligned}E[\hat{\beta}^{\text{R}}] &= E[(X^T X + \lambda I)^{-1} X^T y] \\ &= (X^T X + \lambda I)^{-1} X^T E[y] \\ &= (X^T X + \lambda I)^{-1} X^T X \beta \\ &= \underbrace{(X^T X + \lambda I)^{-1} (X^T X + \lambda I)}_{I} \underbrace{(X^T X + \lambda I - \lambda I)}_{X^T X} X \beta \\ &= (X^T X + \lambda I)^{-1} (X^T X + \lambda I) \beta - (X^T X + \lambda I)^{-1} \lambda I \beta \\ &= \beta - \lambda (X^T X + \lambda I)^{-1} \beta\end{aligned}$$

$$E[\hat{\beta}^{\text{R}}] = X E[\hat{\beta}^{\text{R}}] = X \beta - \lambda X (X^T X + \lambda I)^{-1} \beta$$

$$\|\text{Bias}[X\hat{\beta}^{\text{R}}]\|_2^2 = \lambda^2 \beta^T (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1} \beta$$

$$\begin{aligned}\text{Var}(\hat{\beta}^{\text{R}}) &= \text{Var}((X^T X + \lambda I)^{-1} X^T y) \\ &= \sigma^2 (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1}\end{aligned}$$

$$\begin{aligned}\text{Var}(X\hat{\beta}^{\text{R}}) &= X \text{Var}(\hat{\beta}^{\text{R}}) X^T = \\ &= \sigma^2 X (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1} X^T\end{aligned}$$